Anisa Shaik

anisa.shaik@ufl.edu | +1 352-328-2901| anisashaik.netlify.app/ | linkedin.com/in/anisa-shaik | github.com/anisanazim

EDUCATION

Masters in Computer Science Aug 2024 – May 2026
University of Florida, Gainesville, Florida CGPA – **3.9/4.0**
Coursework*:* Large Language Model, Distributed Computing, Introduction to Data Science, Cloud computing, Big Data Analytics, Mathematics for Intelligent Systems, Artificial Intelligence & Machine Learning, Natural Language Processing

SKILLS

Languages: Python (NumPy, Pandas, PyTorch, Scikit-learn, TensorFlow, SciPy), JavaScript, Java, SQL, C/C++, Julia
Tools & Platforms: Power BI, Tableau, AWS, Docker, Git, Jira
Database Management: SQL, DynamoDB, MySQL, PostgreSQL
ML Techniques: Deep Learning, Natural Language Processing, Reinforcement Learning, OpenCV, NLTK, Generative AI
Development Practices: Agile, SDLC, DevOps, CI/CD Pipelines, Debugging, ADA Compliance

EXPERIENCE

ACIS Lab, University of Florida Gainesville, FL
Graduate Research Assistant May 2025 – present

- Created an **AI agent for iChatBio.org** with **GPT-4**, enabling access to **20M+ occurrence records** from ALA, with **Pydantic + Instructor** ensuring **100% reliable parameters**.
- Architected **modular NLP pipelines** (entity resolution, parameter extraction, response summarization), driving a **15% increase in query accuracy** and a **20% reduction in response time**.
- Leveraged the **iChatBio SDK** to handle A2A protocol abstractions and enhance middleware workflows, contributing to a **25% improvement** in developer efficiency.

Fidelity National Information Services, Inc. (FIS), Chennai Tamil Nadu, India
Software Engineer Jun 2023 – Jul 2024

- Designed and implemented **data validation workflows** for a U.S. government **EBT system**, leveraging **Java/Spring Boot** to process **10K+ daily transactions**, boosting system reliability by **30%**.
- Optimized **SQL queries** and parallelized **batch workflows**, cutting data ingestion latency by **30%** and ensuring **90% system uptime** for mission-critical benefit distribution.
- Automated **bulk card processing and transaction validation workflows** (POS, ATM, e-commerce, voucher, incentive) using loaders and SQL log validation, reducing manual effort by **40%**.
- Enhanced backend **batch processing and settlements** by refining SQL stored procedures and transaction logic, lowering reconciliation mismatches by **25%** and improving **financial reporting accuracy**.

IT Trainee Jun 2022 – Jun 2023

- Integrated **REST/SOAP APIs** to enable secure data exchange between EBT systems reducing manual workflow intervention by 40%.
- **Led QA validation during legacy-to-modern EBT migration**, executing **UAT, regression, and system testing** that reduced post-release defects by **40%** and accelerated production cutover by **2 weeks**.

The Sparks Foundation
Data Science & Business Analytics Intern Sep 2021 – Oct 2021

- Devised a hybrid stock prediction model integrating **LSTM** and **NLP**-driven sentiment analysis achieving 85.52% accuracy (Gradient Boosting) and ~6% improved reliability over single-source approaches.

PROJECTS

Graph Neural Network Recommendation Engine

- Prototyped a movie recommender using **PinSage GCN model**, Improving **Hit@10 by 65%** and **MRR by 47%** over content-based baselines
- Surpassed **traditional methods by 45%** across **25M+ user interactions**, demonstrating scalability for enterprise recommendation systems

Reddit Clone

- Developed a scalable Reddit clone using **Go** and the **Proto Actor** framework, implementing an **actor-based microservice architecture** to handle high concurrency
- Built robust **RESTful API endpoints** with comprehensive error handling, enabling seamless interaction for 100+ concurrent users and achieving **98% response accuracy** in load testing.

Medical MRI Brain Scan Image Segmentation

- **Applied** computer vision techniques to segment brain layers across 1,200+ MRI scans, achieving 98% diagnostic alignment with radiologist annotations
- Employed **U-Net architecture**, improving segmentation performance by 22% compared to baseline models.
- Engineered a hybrid segmentation pipeline combining **active contours**, multi-threshold techniques, and **deep learning models**, achieving **90% reliability** in clinical image analysis